

Distributed restoration in an ATM test bed network

P A Veitch, C O'Shea, D Johnson, P Gaynord, L Nederlof and L Van Hauwermeiren

Network resilience is a crucial area of concern for telecommunications network operators who aim to provide high availability services supported by a broadband platform based on ATM technology. This paper describes the development of an ATM test bed network which incorporates resilience by executing distributed restoration in the event of failures. Since the work originated from collaborative research between BT Laboratories and Alcatel Telecom of Belgium, the paper highlights challenges faced in developing a restorable network, both by a network operator, and a switch manufacturer.

1. Introduction

A variety of options exist for building transport networks to accommodate the wide range of emerging broadband services. The move from plesiochronous to synchronous transmission and the introduction of different types of bit transport services such as switched multimegabit data service (SMDS) and asynchronous transfer mode (ATM), have resulted in the realisation of a multilayer network infrastructure. This raises a number of technical challenges for the future, one of the most significant being to develop a resilience strategy to cover failures within any layer of the transport hierarchy [1—3]. Incorporating resilience into telecommunications networks has become increasingly desirable as customers become very dependent on telecommunications for their day-to-day business needs [4]. The focus of this paper is on ATM network resilience, and describes the development of a laboratory test bed which demonstrates distributed restoration. The role of simulation in validating the exploitation of distributed restoration schemes in larger networks, is also described.

The principles of distributed restoration algorithms (DRAs) are described in section 2, which emphasises the drivers for telcos, and explains the scope for DRAs amid other resilient ATM networking applications. Section 3 describes how a laboratory test bed configuration of ATM switches was developed to demonstrate distributed restoration on real hardware. Section 4 discusses how computer simulations have been used to analyse the scalability of DRAs. Finally, section 5 summarises and concludes the paper.

2. Drivers and considerations for DRAs

2.1 Why DRAs interest telcos

In addition to the bad publicity generated by network outages which seriously degrade quality-of-service, the business community is driving telecommunications operators to provide high availability services, primarily because their revenue depends on it. Furthermore, due to the global liberalisation of the telecommunications market which is now taking place, telcos are actively competing by targeting high availability services in their marketing strategy (Fig 1); added impetus is yielded by the emergence of joint venture networks, correspondent networks, and strategic global alliances which are being formed between operators in different coun-

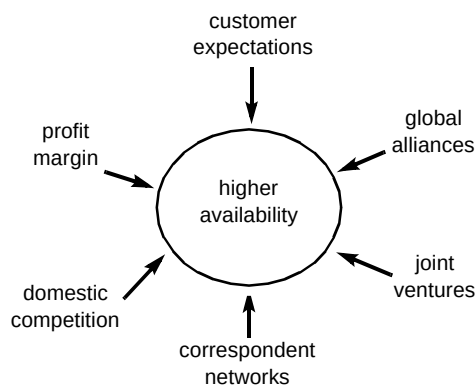


Fig 1 Drivers for better availability.

tries, e.g. Global One and Concert [5]. The quality-of-service and service downtime that an operator can provide becomes a key factor in determining the strength of the role they play in strategic global alliances formed in future years.

Since network maintenance costs can be up to 60 percent of the overall cost of running a national telecommunications network, operators are looking very closely at ways of 'managing' network failures and restoration capacity more effectively to reduce the operational costs and increase profit margins. It is interesting therefore, that one of the principal motivations for exploiting a distributed restoration algorithm (DRA) in place of centralised restoration, is that it does not require a network management system (NMS) to be involved in the restoration process [6]. Instead, messages are passed between the nodes in the network to build up a picture of the connectivity at the time of failure. In this way, re-route paths can be found in real time without the requirement for routeing tables that need to be continuously updated, meaning maintenance costs are lowered.

2.2 Building resilient ATM networks with DRAs

ATM has a different meaning to different people. Some see it as a new data networking technology, whereas others see it as a new infrastructure which will unify the transport of telecommunications services. It is perhaps not surprising therefore, that a whole range of resilience mechanisms have been proposed for ATM, some drawing on ideas from the data networking world, while others come from transport networks.

P-NNI (private network node interface) specifies a dynamic routeing and restoration capability similar to dynamic routeing principles used in IP (Internet protocol) networks [7]. It requires each network node to maintain a local database of information on the topology of its own 'peer group' plus information on how to route to other groups. To route or restore a circuit, the source node calculates a route from its local data and initiates a signalling sequence across the network. Clearly, this relies on the local database being up to date. In the event of a network failure, there will be a time lag before all nodes are aware of the new topology. In the meantime, restoration attempts may be routed towards the site of the failure, so a crank-back mechanism has also been specified, but this will yield less optimal routeing.

A cable break in the wide area network may affect thousands of connections simultaneously, so jeopardising the potential for fast restoration. This is why mechanisms have been proposed to operate at the virtual path (VP) level, in an attempt to restore bundles of connections at the same time. A variety of schemes have been proposed which involve predetermining the route which will be used as a

back-up to the usual working VP route [8, 9]. Some call for bandwidth to be reserved on the back-up route, resulting in dedicated 1+1 protection, and allowing the signal to be broadcast on both routes while using a receive-end switch. Others allow the back-up bandwidth to be shared, using less transport capacity, but requiring a simple switching protocol and a check that the bandwidth is still available when it is required. These methods do not require the network nodes to have any network knowledge, simply an alternative entry in the routeing table for the back-up route.

Work on the test bed has focused on a third group of resilience mechanisms known as distributed restoration algorithms (DRAs). The concept was first proposed for use in SDH or SONET transport networks [10], but is equally applicable to ATM. It combines two attractive features of the previous mechanisms — there is no need to maintain a network database in each node, yet at the same time it does not rely on a pre-planned back-up path which may be out of date. Instead, a dynamic distributed routeing algorithm is used to search the network at the time of the failure, so the network acts as its own routeing database and cannot be out of date. Other features of the distributed restoration technique are:

- it uses less network resources than dedicated protection switching,
- it is faster than centralised restoration,
- validation of network restorability is possible,
- it can be run off-line and then executed under manual control,
- it can provide post-restoration log files which can be interrogated via the network manager (for back-to-normal operation, etc).

It is unlikely that dynamic distributed restoration will be deployed by a telco as the only means of providing restoration within a multilayer network infrastructure. For example, the DRA could be used as part of a resilience strategy where pre-planned back-up paths are first used [3]. By escalating to dynamic distributed restoration, it will be possible to search out any additional spare capacity in the network and hence provide additional candidate re-route paths that the pre-planned routeing table does not have.

2.3 Span-based versus path-based DRAs

Two main variants of DRA have been investigated as shown in Fig 2. One is a span-based DRA which restores traffic between the ATM crossconnects adjacent to a failed network span (e.g. a failed line system or severed cable). The other is a path-based DRA which restores traffic

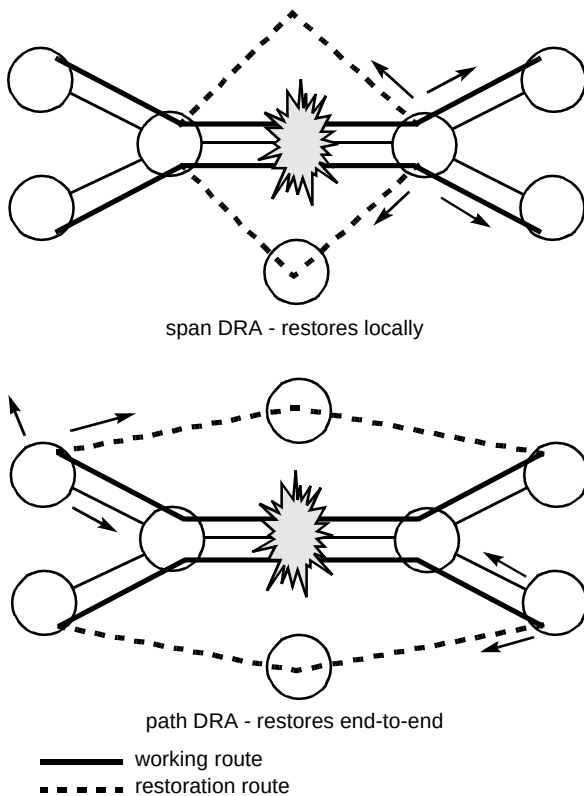


Fig 2 Operation of span and path DRA.

between the source and sink of the affected trials regardless of where the fault occurs. Both operate at the VP level.

The span DRA is simpler when considering a single span failure because the path DRA will result in multiple pairs of source-sink nodes trying to reconnect while competing for spare capacity in the network. However, once the necessary contention-handling mechanism has been introduced, the path DRA is inherently able to handle multiple span and node failures, while the span DRA requires further extension. This is important because multiple span failures are actually quite common in real networks. Consider Fig 3 for example, where a single cable cut results in three logical span failures as a result of line systems bypassing some switches to reduce costs.

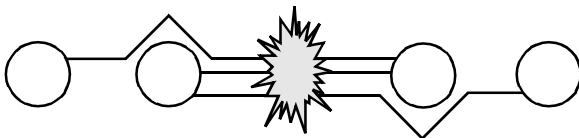


Fig 3 Single cable failure resulting in multiple span failures.

It is not possible to say that either the span or the path DRA is inherently faster, even for a single span cut. The span DRA is simpler, but the nodes adjacent to the failed span act as bottle-necks, since each must process messages for every failed connection and also switch every failed connection. In the path DRA, no single node will have this

burden, but each may have to process messages from many simultaneous restoration problems.

A key factor in favour of a path DRA is that it makes better use of spare capacity in the network and can get by with 30—40% less spare capacity in a typical network [1]. However, the path DRA is less predictable in the routes it will find, so it may be necessary to sacrifice some of this saving if 100% restoration is required.

3. Integration of a span DRA in a laboratory test bed

3.1 DRA implementation considerations for vendors

The introduction of distributed restoration algorithms into transport networks imposes additional requirements on the network elements. During the restoration process, messages will be flooded in the network between the instances of the restoration algorithm located in neighbouring network elements. The network elements need to be flexible in order to provide this inter-element communication channel. Because the objective is to minimise the network outage time, the message transfer delays between, and integral to, network elements should be minimised. Hence, it is important for performance reasons to carefully locate the restoration algorithm in the element architecture, because of various interactions that the algorithm has with the element resource manager, alarm handling functions, message transport, network management, etc. This raises the question of which interfaces need to be open (and standardised) if the restoration algorithm is seen as a function which can be added into the network in a modular way — an issue which is even more important if the network consists of multivendor equipment. It follows that in order to enhance an operational network which supports distributed restoration, it is necessary to introduce algorithms into element architectures for which they were not originally designed.

Adding a restoration algorithm into a network element also raises the issue of having two managing entities for a single network element. In addition to the interface to the classical network manager, a control interface from the restoration algorithm host to the element resources must be introduced. The restoration function will instigate the setting up or clearing down of connections (or parts of connections) in the network element, and will report these changes to the network manager afterwards. Therefore, the design of the network element architecture should be capable of coping with these (sometimes conflicting) management functions.

3.2 Test bed description

At BT Laboratories (BTL), and in the research laboratory of Alcatel Telecom, a test bed has been set up for demonstration of a distributed span restoration algorithm. This algorithm, developed by Alcatel [11], was previously

validated by extensive simulations on a number of network topologies of up to 32 nodes [12, 13].

Clearly, the DRA works fastest when it is implemented close to the transport layer, which avoids time-consuming protocol stacks in the communication between software and hardware components inside a network element. The DRA was implemented in software for several reasons. Firstly, modifying hardware would be technically difficult and time-consuming during the lifetime of the project. Secondly, a software-only implementation offered more flexibility for experimentation. The DRA process was installed close to the network management interface, since the required functions (such as setting up or deleting connections) and information (working and spare resources) are, in principle, available at that point.

Since the intention was to demonstrate a DRA as realistically as possible, a commercially available Alcatel switch was used. In the modular structure of the system, only the link termination board (TLK) was used as a single-board 8 × 8 ATM crossconnect, with a full software structure, albeit with some modifications to incorporate the absence of switching planes and rack-level control boards. This software runs on a control workstation (CWS) that is connected to the TLK. Initially five, and later seven nodes (TLK + CWS), as well as a number of optical fibres, were available for the demonstrations and experiments.

The general structure of the software is given in Fig 4, which illustrates some necessary changes that were made for the management of the DRA in the test bed, as well as an ATM adaptation layer (AAL5) process that was added to the on-board controller (OBC) of the TLK to accommodate DRA messaging between neighbouring nodes. The software module that contains the core code of the algorithm was directly imported from the simulation platform on which the DRA was originally developed and validated. This, as a by-product, offered valuable opportunities to study and compare the behaviour of the algorithm in a simulated environment, with that of a real environment.

Although a DRA can operate autonomously without interaction with the network manager, applying a management platform in the test bed still proved to be very useful. Some applications were developed for managing the DRA and several extensions were made to the management information base (MIB). In the initial implementation, this management interaction was not the prime focus of the experiments, but nevertheless added value to the demonstrations by a graphical display of events during the restoration process. The continuation of this project as a collaboration between BTL and Alcatel, will involve the study of management aspects of distributed restoration, both from a network operator's and a manufacturer's viewpoint. Also, while a span DRA has been the focus of the

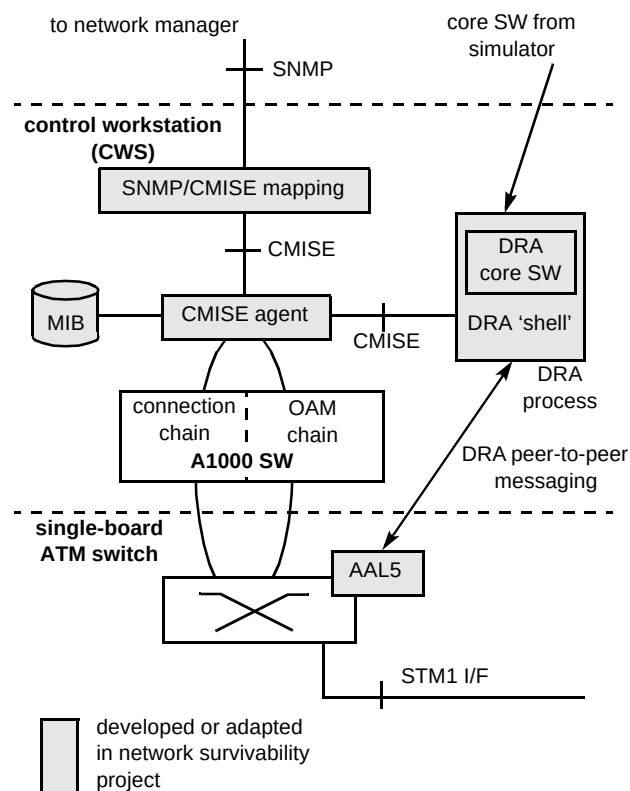


Fig 4 Structure of ATM crossconnection tested.

experiments thus far, a path DRA is being developed at the test bed situated at BTL.

3.3 Results

To date, the main objective of the test bed has been to demonstrate distributed restoration functionality; performance improvements will be addressed next. As a result, although the performance figures given here exceed the 2-second call disconnection objective [6], the experiments have been an invaluable means with which to identify the main bottle-necks from which guidelines for future products can be formulated.

Most of the demonstrations have been given with a single VP semi-permanent connection (SPC) carrying a video-on-demand (VoD) application, with a VoD server and a VoD user connected to different nodes in a 5-node network. After simulating a link failure by disconnecting a fibre, the video image was usually restored 15–20 seconds later. More extensive tests have been performed on the 7-node network shown in Fig 5, with a traffic generator generating and receiving 16 VP SPCs. The link between node 4 and node 5 was cut and spare capacity in the network was dimensioned such that the DRA was forced to restore the connections on 5 different alternative routes, as shown in Fig 5. The resulting restoration times are plotted in Fig 6

which shows that the failed connections were completely restored 1 to 2 minutes after the link cut.

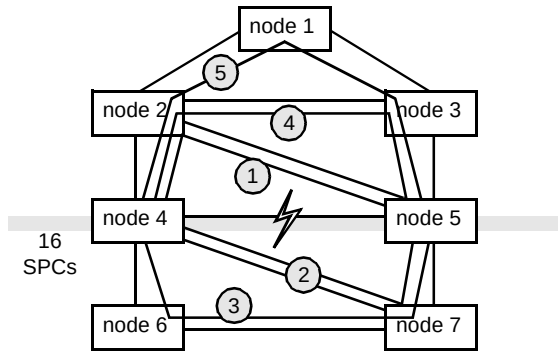


Fig 5 Testbed re-routing.

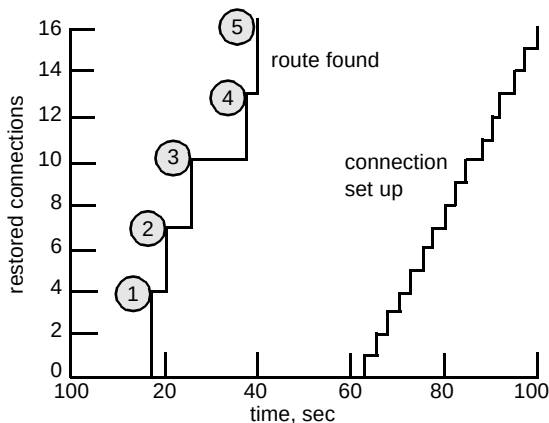


Fig 6 Time results of restoration in laboratory test bed.

The total time needed for restoration can be split into three parts:

- alarm detection time (not indicated in Fig 6) — this is the time that passes between the cutting of the fibre and the activation of the DRA, usually 3—5 seconds,
- DRA execution time (left line in Fig 6) — this is the time that the DRA takes to find alternative routes with spare capacity, and to allocate failed connections to those routes,
- time needed to set up the connections in all nodes (right line in Fig 6) — this is the time needed to complete the creation of the connections in all nodes that are involved in the re-routing.

The execution time of the DRA is mostly a result of the message transfer delays. The bottle-neck here is the AAL5 process that runs on the on-board controller of the TLK, and handles both incoming and outgoing messages. The resulting message transfer delay depends on the current load of the on-board controller and on the length of the DRA message, but rapidly runs into the seconds time-scale. A major increase in performance can be achieved by improving the AAL5 handling process, for instance by connecting the control workstation with an ATM link to the

node. This reduces the message transfer delay to several milliseconds. Simulations using these values have shown that this phase of the restoration will then take between 0.5 and 0.8 seconds. A second, significantly large delay is the time required to set up the connections once the DRA has determined the new routes. This is as a direct result of the limited functionality offered by the crossconnect software. The only way for the DRA to set up a connection is to pass through the local element manager, each call taking several seconds to complete. In addition, the nodes that terminate the failed span — nodes 4 and 5 in Fig 5 — initially have to delete the original connections explicitly which requires another two calls to the element manager. It is evident that the connection set-up time can be greatly reduced by giving the DRA direct access to the routing tables.

4. Simulating distributed restoration algorithms

One of the main benefits of conducting experiments on a test bed is understanding practical constraints and implementation problems encountered which are not necessarily incorporated into an abstract simulation model. The knowledge gleaned from test bed experimentation can in turn be fed back into a computer simulation, since more realistic processing architectures can be modelled, thus providing more credible results. The principal restriction with test bed experimentation, however, is the size of the network being constructed, which is constrained by costs and implementation overheads. Simulation tools have therefore been exploited as a means to evaluate systems such as large self-healing telecommunications networks, which would be otherwise impractical to construct in the form of a test bed. At BTL, simulation has been employed for DRAs in both SDH and ATM networks, including a multilayer network model which demonstrates the principle of escalation between layers in the event of a cable break [14].

The simulation tool enables a modular and object-oriented model of a network to be specified hierarchically, thus reflecting real communications systems. Given the physical and logical network definitions, performance modelling of some chosen aspect of the network functionality can be undertaken — the main interest is in verifying the functionality of distributed restoration, and analysing relevant performance metrics such as restoration time. From Fig 7, the interconnection of nodes with links is defined at the network level, which represents how the ATM network elements (such as the Alcatel switch detailed in the previous section), are connected together by fibre transmission systems. At the node level, the internal architecture of the ATM switch can be defined. It is here that the necessary abstractions may be made to produce an accurate, yet manageable representation of the structure of the network element. For example, since the performance of a DRA relies principally on the processing and communication of flooding messages, modelling at the node level concentrated on the extraction, processing, generation, and re-insertion characteristics of such messages within a

switching node's architecture. The lowest level of the modelling hierarchy is the process level, where the actual DRA functionality is embedded into the overall network model. Both the Alcatel span DRA, and a path DRA developed at BT, have been investigated by simulations.

The scalability of employing DRAs to restore ATM circuits was confirmed by simulating a 32-node network (Fig 8) with 340 VPs. The DRA messages were assumed to be 64 bits long which take 10 ms to be processed within nodes, and are transmitted between nodes at 64 kbit/s. The actual VP crossconnection time was assumed to be 20 ms. The results of simulating the BT path DRA are plotted in Fig 9 which demonstrate 100% restoration of VPs in around 7 seconds. Such results are very useful in demonstrating what can be achieved if certain node processing times are sufficiently reduced. The goal of future test bed development is to continuously reduce the message processing overhead in an effort to achieve restoration times of the order of a few seconds.

5. Conclusions

This paper describes the collaboration between research centres of BT and Alcatel Telecom, which has led to the development of a working laboratory test bed of 7 ATM

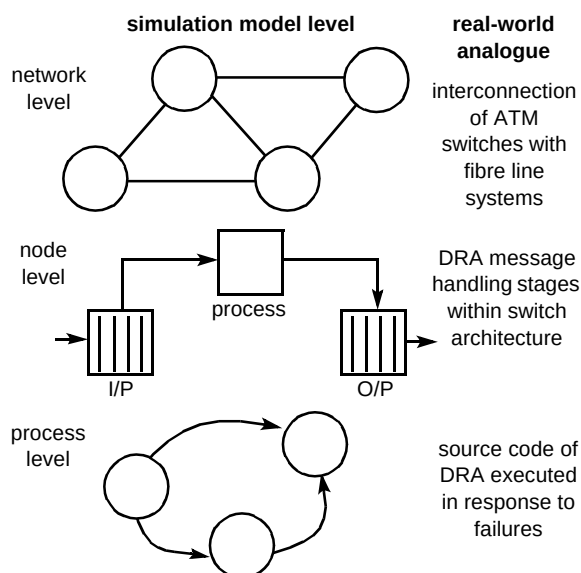


Fig 7 Relating the simulation model to actual telecoms networks.

nodes which successfully execute a distributed restoration algorithm (DRA) in response to fibre failures. Having outlined the drivers for a telco to exploit DRAs, and the advantages and disadvantages of different types of DRA, the approach taken by Alcatel to adapt the architecture of a commercially available switch into a prototype for

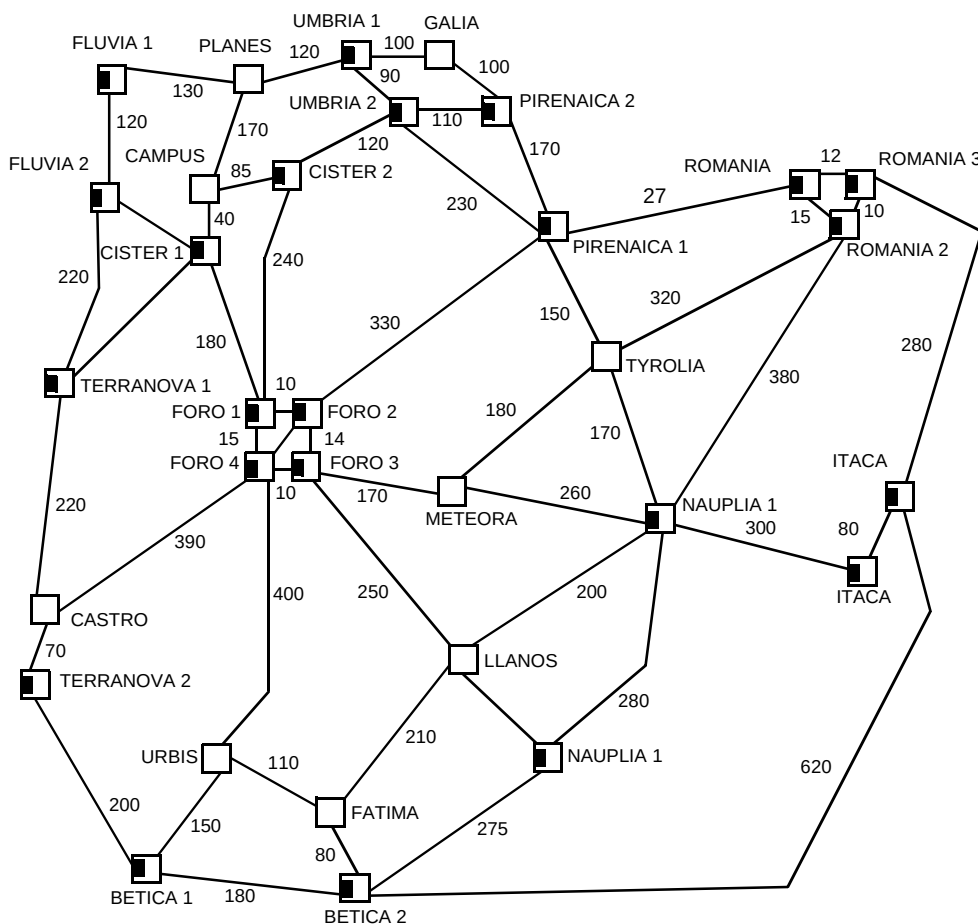


Fig 8 32-node test network for DRA simulations.

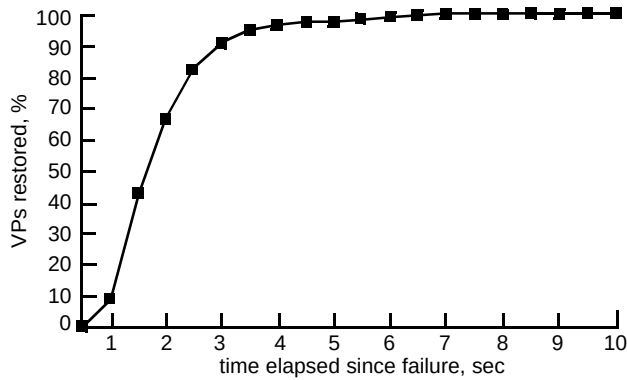


Fig 9 Cumulative restoration (%) resulting from path DRA simulation.

supporting DRA functionality, has been explained. Some important considerations which impinge on the adaptation of the switch architecture are:

- location of the restoration software in the architecture, and the necessary interactions between it and resource management, fault management, and message transport handling modules,
- efficient formation of DRA messages by using ATM adaptation layer (AAL) processing in the switch,
- communication required between DRA software and the network management interface.

All of these matters were addressed in the development of the test bed apparatus which was used successfully to demonstrate the restoration of a video-on-demand application in 15 seconds. Further testing of the restoration of several connections at the same time was undertaken which enabled items causing the slow response in the DRA processing to be identified, the main bottle-neck being AAL message handling. However, the slow response could be improved by using faster links in the intra-nodal communications network. Scalability of DRAs on larger networks beyond the magnitude of the test bed was confirmed by computer simulations of a 32-node network topology. By combining the joint expertise of BTL and Alcatel Telecom in the fields of network simulation and switch design, the viability of DRAs for ATM network restoration has been demonstrated.

References

- 1 Johnson D: 'Survivability strategies for broadband networks', IEEE Globecom '96, London, pp 452—456 (November 1996).
- 2 Johnson D et al: 'Distributed restoration in telecommunications networks', BT Technol J, 12, No 2, pp 67—76 (April 1994).
- 3 Nederlof L, Struyve K, O'Shea C, Misser H, Du Y and Tamayo B: 'End-to-end survivable broadband networks', IEEE Communications Magazine, 33, No 9, pp 63—70 (September 1995).
- 4 McDonald J C: 'Public network integrity — avoiding a crisis in trust', IEEE J-SAC, 12, No 1, pp 5—12 (1994).

- 5 'Telecoms Markets', Issues 294 and 305 (1996).
- 6 Wu T H: 'Fiber network service survivability', Artech House (1992).
- 7 ATM Forum: 'P-NNI V1.0', spec of af-pnni-0055.000. (<http://www.atmforum.com/atmforum/specs/approved-specs.html>)
- 8 Kawamura R, Sato K I and Tokizawa I: 'Self-healing ATM networks based on virtual path concept', IEEE J-SAC, 12, No 1, pp 120—127 (1994).
- 9 Veitch P A, Hawker I and Smith D G: 'Administration of restorable virtual path mesh networks', IEEE Communications Magazine, 34, No 12, pp 96—101 (1996).
- 10 Grover W D: 'The self-healing network: a fast distributed restoration technique for networks using digital cross-connect machines', Globecom '87, Tokyo, Japan, pp 1090—1095 (1987).
- 11 Nederlof L, Vanderstraeten H and Vankwikelberge P: 'A new distributed restoration algorithm to protect ATM meshed networks against link and node failures', ISS '95, pp 398—402, Berlin, Germany (April 1995).
- 12 Ocakoglu G, Nederlof L and Hauwermeiren V: 'ATM meshed networks, protected by distributed restoration mechanisms', 13th Annual Conference on European Fibre Optic Communications and Networks, pp 102—105, Brighton, England (June 1995).
- 13 Vanderstraeten H, Chantrain D, Ocakoglu G, Nederlof L and Van Hauwermeiren L: 'Integration of distributed restoration procedures in the control architecture of ATM crossconnects', Broadband Communications '96, pp 549—560, Montreal, Canada (April 1996).
- 14 Veitch P A, Chng R S K, Johnson D, Hawker I and Smith D G: 'An integrated restoration system for an SDH-based ATM transport network', IEEE Globecom '96, London, pp 1882—1886 (November 1996).



Paul Veitch received MEng and PhD degrees in Electronic and Electrical Engineering from the University of Strathclyde in 1993 and 1996.

Upon completion of a PhD thesis detailing ATM network restoration techniques, he joined BT and has since been involved in the design and analysis of resilient transport network infrastructures.



Chris O'Shea graduated with a BSc in Electronics from Salford University in 1981. He joined the Optical Submarine Transmission unit of BT in the same year. From 1981 to 1992 he was part of a team that developed the first operational transatlantic optical fibre transmission system. During this period, he also studied part-time for an MSc in Telecommunication Systems at Essex University. In 1992 he moved to another unit within BT Laboratories involved in the planning and design of broadband data and transmission networks. He has been actively involved in developing planning and

resilience tools to support the roll-out of SDH and ATM-based networks.



Dave Johnson joined BT in 1986 after graduating from Lancaster University with a first class honours degree in Physical Electronic Engineering. He worked for five years on systems reliability, including the reliability analysis of network architectures and the production of computer software for system reliability prediction. He then spent six months working on intelligent networks. He currently leads a small team in the Network Transport Engineering Centre. He is involved in transport resilience of existing networks, and the development of a strategy to integrate the resilience mechanisms

available in a range of network technologies to meet customer requirements in the most cost-effective manner.



Peter Gaynord graduated with first class honours in Combined Sciences in 1986. He joined International Computers Limited as a Software Engineer and worked on the development of advanced artificial intelligence applications for design to manufacture of electronic products. In 1990 he moved to Schlumberger Cambridge Research where he developed a commercial gas kick simulator for use in oil field blow-out prevention.

In 1994 he established Gaynord Consulting Limited and has since worked exclusively for the broadband and data networks group within Networks and Systems at BT Laboratories. He has developed strategic planning tools for designing network resilience for SDH, ATM and PDH networks. Recently this work has included the development of an ATM path-based distributed restoration algorithm for test bed demonstration and testing.



Leo Nederlof received an MScEE from the Delft University of Technology in the Netherlands in 1992, and joined the Alcatel Research Division in Antwerp, Belgium, in that same year. He has been working on network survivability and self-healing networks, and contributed to the RACE 1022 project and the RACE 2101 project IMMUNE, and is currently contributing to the ACTS project AC208 REFORM. He took part in the development and validation of a new distributed restoration algorithm, which was successfully implemented and demonstrated in an ATM laboratory test bed network. Now

leading the Network Survivability activities of the Alcatel Telecom Research Division, he is taking part in a project on survivable all-optical network architectures and is contributing to standardisation of survivability in ATM, SDH and WDM optical networks.



Luc Van Hauwermeiren received the MScEE degree from the University of Gent (Belgium) in 1987. He joined the research team of the Department of Information Technology at the University of Gent, where he was involved in the research on modelling and characterisation of high-speed interconnections.

Since 1992 onwards, he has been associated with the Research Division of Alcatel Telecom, where he has been involved in the system definition and specification of broadband network technologies. In particular, he has been responsible for research projects on network resilience and broadband internetworking.

He recently joined the Access Systems Division of Alcatel, where he is currently involved with the product management of the A1000 ADSL product line.

